



David M. Everly Jr

Software Systems Engineer • Acute Care Nurse Practitioner

MSN, CRNP, AGACNP-BC
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PROFESSIONAL SUMMARY

I am a Software Systems Engineer and Acute Care Nurse Practitioner with over a decade of experience in Critical Care and Emergency Medicine. I have extensive experience operating under pressure in environments where risk is real and facts are ambiguous. I understand risk, trust, and safety and leverage this to design practical software systems that are intentionally built to streamline workflows, minimize risk, and mitigate the impact of system failures.

SYSTEMS & TECHNICAL FOCUS

Secure system design • defensive architecture • zero trust • adversarial input handling • applied machine learning • workflow automation • reliability under constraint

SELECTED ENGINEERING PROJECTS

SynapSys — Failure-Aware LLM Orchestration Framework

- Designed modular orchestration layer enforcing validation, guardrails, and bounded repair
- Explicitly models failure modes, misuse, and controlled degradation around probabilistic output
- Enforced input and output validation layers with deterministic fallback behavior

Portfolio Chatbot — SynapSys Reference Implementation

- Built retrieval-augmented LLM system with pre/post guard pipelines and deterministic fallback
- Demonstrates trust-boundary enforcement and guarded interaction patterns

Additional Engineering Work

- Built ECG classification models from raw waveform data with class imbalance normalization
- Developed constraint-based scheduling automation system using simulated annealing
- Practical study of network segmentation, service isolation, and basic adversarial security testing

CLINICAL & OPERATIONAL EXPERIENCE

WellSpan Pulmonology & Critical Care Medicine — Nurse Practitioner 2019–Present

- Manage high-acuity patients under time pressure, ambiguity, and evolving conditions
- Perform advanced procedures and lead real-time clinical decision-making
- Identify workflow, documentation, and system failure points impacting safety and efficiency
- Preceptor for advanced practice clinicians and trainees

Hershey Medical Center Emergency Medicine — Nurse Practitioner 2017–2019

- Emergency medicine practice in high-volume, safety-critical settings

Registered Nursing Roles 2013–2017

- Progressive care, emergency medicine, cardiopulmonary step-down, clinical instruction

EDUCATION

M.S. Computer Science (Anticipated May 2026) — Drexel University 2022–2026

Graduate Certificate, Computational AI — Drexel University 2024–2025

M.S. Nursing — The Pennsylvania State University 2015–2017

CREDENTIALS & RECOGNITION

CompTIA Security+ (exam pending) 2026

ISC2 Cybersecurity Certified 2025

Epic Physician Builder 2024

AGACNP-BC 2017

Registered Nurse 2013

Advanced Practice Provider of the Year — Hershey Medical Center 2019

Graduate Fellowship — The Pennsylvania State University 2023

ENRS Abstract 2017